

APACHE COUCHDB: It's Time to Relax



Apache CouchDB frees developers from restrictions caused by the uniformity and structured nature of relational database management systems. Tailored for the needs of the data driven Web applications, CouchDB is definitely a promising data manipulation tool for Web developers.

The ability to store data persistently and to retrieve it when required is the most important aspect of software applications. There are various methods to store data persistently—files, databases, etc. Flat files allow the storage of data, though retrieving them using filters is not a sophisticated option. The idea of databases came into existence to address the issue of data storage and retrieval. There are different types of database management systems, which include:

- Relational data model
- Hierarchical data model
- Network data model
- Object based data model, etc.

Relational database management systems (RDBMS) have dominated the other types mentioned above due to various advantages such as tabular structure, linked data, integrity and the ability to apply a standardised, uniform language SQL to retrieve the relevant data in a relatively effortless manner. The ability to satisfy the ACID (atomic, consistent, isolated and durable) properties is one of the key features of an RDBMS. However, the structured nature of the RDBMS, which is normally considered as an advantage, can be seen as a restriction as well, when mapping hybrid real-world data.

NoSQL

NoSQL has evolved to make a significant shift from the traditional RDBMS approach. NoSQL refers to databases

that don't need to follow a fixed schema, and means 'Not Only SQL'. It is primarily designed as a distributed data store, which is well suited for Web scale applications. NoSQL stores the data as key-value pairs rather than following a rigid predefined schema, and is focused on the CAP theorem (consistency, availability and partition tolerance). NoSQL data stores are focused on high availability, scalability and high performance. These databases come under the following categories:

- Key-value pairs
- Column-oriented
- Graph based
- Document based

There are many database tools belonging to the NoSQL type as illustrated in Figure 1.

CouchDB

It can be observed from Figure 1 that Apache CouchDB is classified as a NoSQL database under the document based category. The development of CouchDB, by Damien Katz, dates back to 2005. CouchDB is managed by the Apache Software Foundation. The code for it is written in the Erlang programming language. CouchDB facilitates the storage of data as documents that don't need to be restricted by a rigid structure. The document based data stored in CouchDB are JSON documents. HTTP is used as an interfacing method with CouchDB.